DISSENT

The machine second opinion on America’s bridges

TEAM NEXUS NETWORK

EVERY RATED BRIDGE IN THE UNITED STATES · 621,137 STRUCTURES · 41,319 RATED POOR

DEPT. OF CSE (CYBER SECURITY)

THE PROBLEM

A bridge does not fail the day it becomes unsafe.

It fails the day the paperwork and the physical bridge stop agreeing — and nobody is looking at both.

24 months between routine inspections, so a defect born the day after one waits a year on average to be seen

0–9 a coarse, inspector-subjective condition scale is the entire official record of a structure

< 20% of the world’s long-span bridges carry any monitoring hardware at all

THE FORENSIC RECORD

In every collapse, the warning already existed.

Eight documented failures across three continents. In each one the evidence was on file, in time to act, and belonged to nobody.

Morandi Bridge

GENOA, ITALY · 14 AUG 2018 · 43 DEAD

A satellite radar scatterer on the deck beside the failed pier accelerated from about 10 to 70 mm/yr starting March 2017, seventeen months before collapse (Milillo e...

I-35W Mississippi River Bridge

MINNEAPOLIS, USA · 1 AUG 2007 · 13 DEAD

Inspection photographs from 1999 and 2003 already showed visibly bowed gusset plates (NTSB HAR-08/03).

FIU pedestrian bridge

MIAMI, USA · 15 MAR 2018 · 6 DEAD

Rapidly growing structural cracks, photographed and discussed by the engineer of record in the days before failure.

Champlain Towers South

SURFSIDE, USA · 24 JUN 2021 · 98 DEAD

A satellite study published in 2020 had identified the site subsiding by about 2 mm/yr in the 1990s (Fiaschi and Wdowinski); a 2018 engineering report documented maj...

Morbi suspension footbridge

GUJARAT, INDIA · 30 OCT 2022 · 135 DEAD

Reopened four days after a renovation that never replaced or load-tested the original main cables; the inquiry later found 22 of the 49 wires in the failed cable alr...

Carola Bridge

DRESDEN, GERMANY · 11 SEP 2024

Hidden hydrogen-induced corrosion of the prestressing steel, progressing for decades inside the one deck not yet refurbished (Marx report, TU Dresden).

THE IDEA

Infrastructure does not fail silently. It fails contradicted.

DISSENT keeps two independent accounts of every structure and files their disagreement.

THE PAPER WITNESS

What the institution believes

Condition ratings 0–9, thirty-four years of filings, maintenance and construction history — every human judgement quarantined here.

THE PHYSICS WITNESS

What the evidence shows

Age, traffic and truck load, structural form, and real freeze–thaw and flood exposure. It is never shown a single inspector’s opinion.

When the two stop agreeing, the machine files a dissent: a dated obligation with the evidence attached.

HOW IT WORKS

From the federal record to a dated obligation.

01 LEDGER

234,801 inspection filings across 4 states parsed into one trajectory per structure.

02 EVIDENCE

Physical features joined per asset: age, works history, traffic, form, and real ERA5 weather exposure.

03 RE-INSPECT

A gradient-boosted model predicts the rating from physics alone. Trained only to 2015; everything after is unseen.

04 CALIBRATE

Split-conformal intervals: every verdict ships with ±1.4 rating steps of honest uncertainty.

05 DISSENT

Three channels: the record claims better than evidence; the gap is accelerating; physics alone says the asset is severe.

06 DOCKET

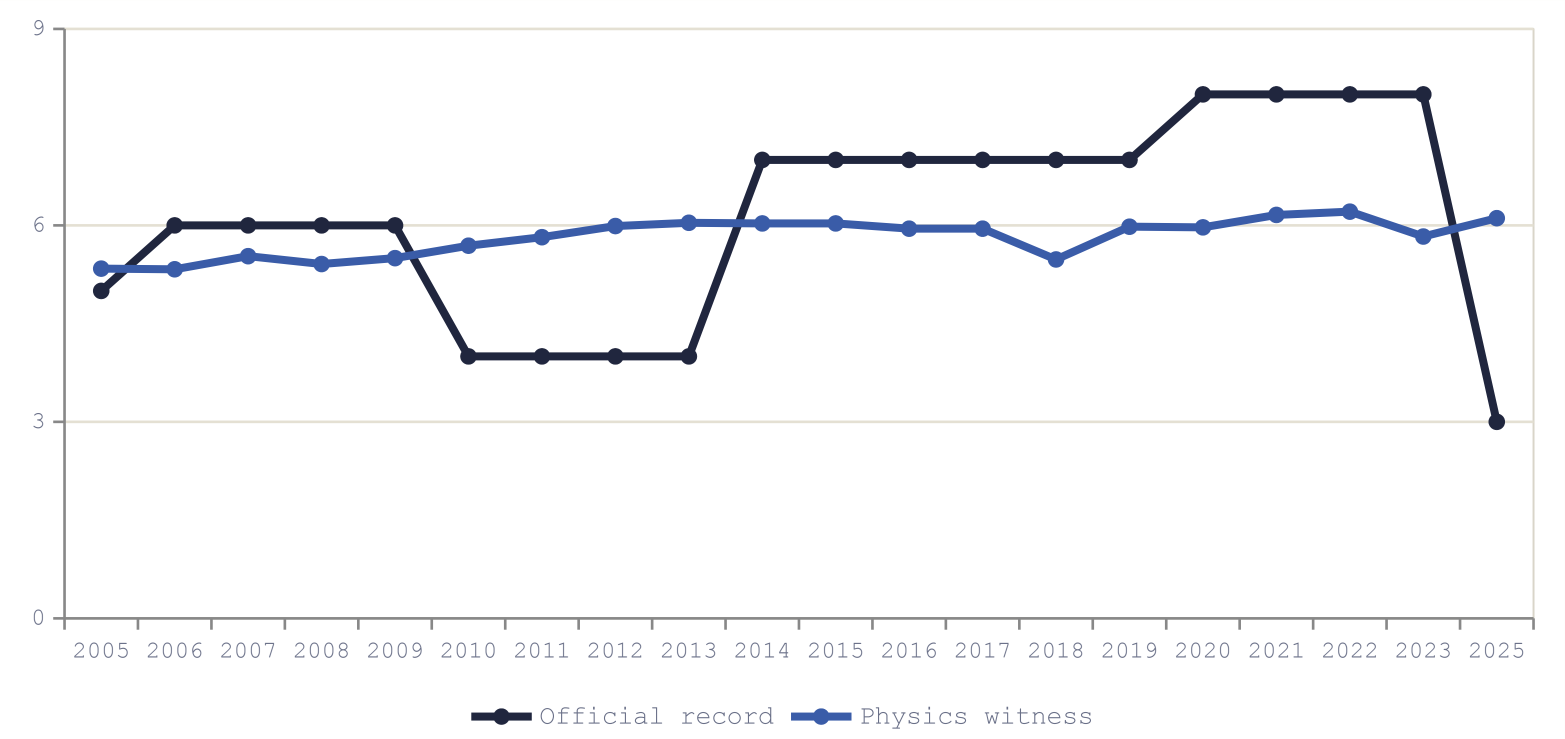
Capped to real capacity — 12 mandatory, 24 scheduled, 48 watched per quarter.

No installed hardware. Every input is free and public.

THE PROOF

Flagged 9 years before the record caught up.

Vermont structure 101412003714121: the official rating fell from 8 to 3 in 2025. The model, frozen at 2015 and blind to everything after, had it inside the top-15% alert budget 9 years earlier — and kept it there every year since.



9 yrs

LEAD TIME BEFORE THE RECORD MOVED

8 → 3

WHAT THE RECORD FINALLY FILED

2025

YEAR THE PAPERWORK CAUGHT UP

Real federal data. The model never saw a single post-2018 filing.

VALIDATION ON THE FUTURE

We hid the answers, then checked.

1,835 "the record was forced to catch up" events were mined from the trajectories: sudden drops of two or more rating steps, or closures. The 166 that happen after 2018 are pure holdout — the model was frozen in 2015 and never saw them.

234,801

REAL INSPECTION FILINGS AUDITED

0.7

RATING STEPS OF ERROR ON UNSEEN YEARS

89%

CONFORMAL COVERAGE (TARGET 90%, REPORTED HONESTLY)

23%

OF HELD-OUT FAILURES CAUGHT EARLY
1.5× BETTER THAN CHANCE

3.5 yrs

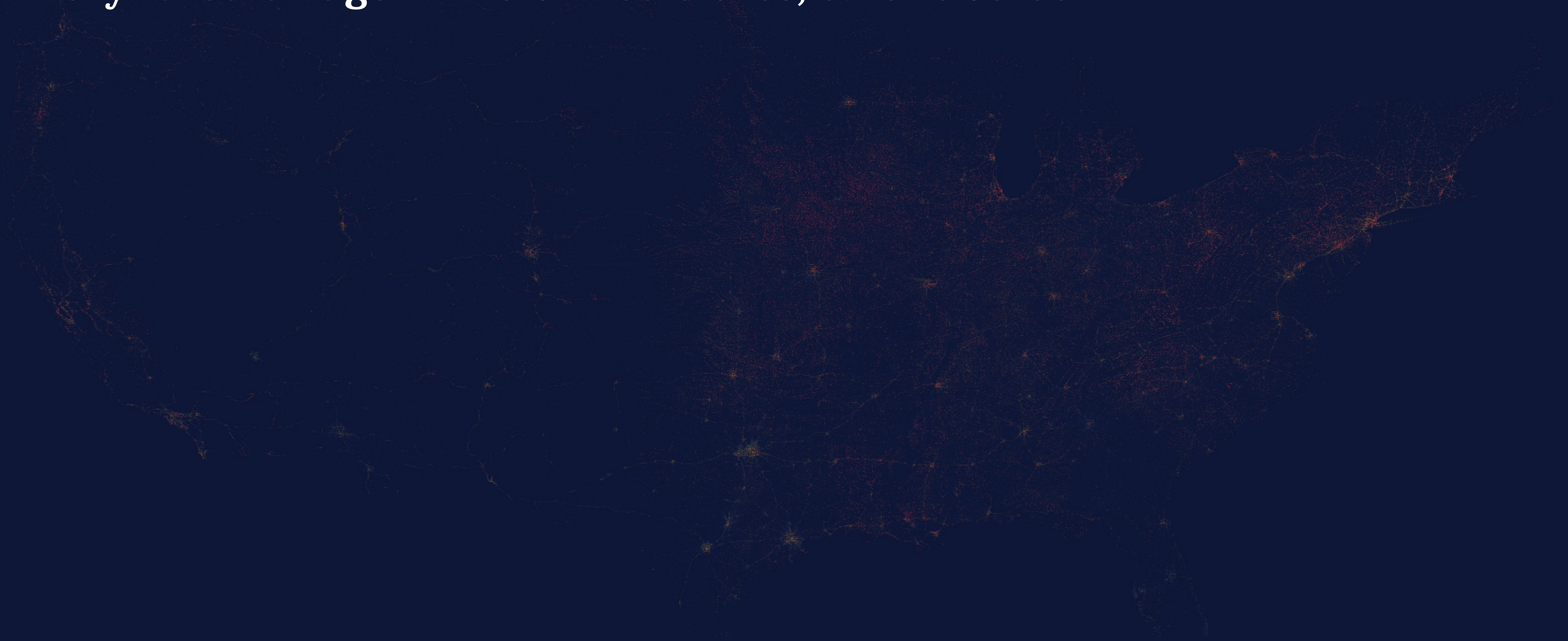
MEDIAN WARNING BEFORE THE RECORD MOVED

Why an alert budget, not an accuracy score

A model that flags everything catches everything and helps nobody. Ours is scored the way an inspection team actually works: a fixed number of alerts per quarter, and we count only the real failures that fall inside it.

SCALE

Every rated bridge in the United States, on one screen.



621,137
STRUCTURES PLOTTED

41,319
RATED POOR

9,765
AUDITED IN DEPTH

9.9 MB
NO BACKEND REQUIRED

WHY THIS IS NEW

Everything else waits for one signal to get loud.

APPROACH	WHAT IT DOES	
Sensor monitoring	Hardware on one bridge, alarms on a threshold	Thousands per channel; almost nothing is instrumented
Satellite screening	Millimetre ground movement, delivered as reports	One modality, and no owner for the finding
Risk models	Rank cohorts by age and material	Cannot see an individual asset drift between inspections
Inspection AI	Computer vision on inspection photographs	Inherits the inspector's verdict as ground truth

DISSENT

Audits the record itself, on 100% of an inventory, with zero installed hardware — and its output is a calibrated contradiction with a dated obligation attached, not another dashboard.

WHAT IT CANNOT DO

The limits, stated before you ask.

It abstains rather than guess

Structures newer than five years sit outside the model’s training support, so it issues no verdict on them at all — 81 of them in Rhode Island alone.

Coverage is reported, not tuned

Our conformal intervals cover 89% of unseen cases against a 90% target. We report the shortfall rather than retune on the test years.

Some failures have no precursor

Sudden scour in a flood, or a construction-phase error on a structure too young to have a record, is invisible to any method built on filings.

The satellite channel is not in this pilot

No free processed radar covers these states, so the replay demonstrates that detector on a published record instead — and cites the paper that disputes it.

A tool that audits records has to be auditable itself.

WHY IT MATTERS

A fixed inspection workforce, pointed at the right bridges.

730 days

TODAY: THE AVERAGE GAP BETWEEN LOOKS AT A STRUCTURE

6–12 days

WITH DISSENT: THE CADENCE OF A FREE SATELLITE REVISIT

≈ $0

MARGINAL COST PER STRUCTURE: EVERY INPUT IS PUBLIC

A county with five bridges and one engineer gets the same second opinion as a national railway, because there is nothing to install and nothing to buy.

NEXT JURISDICTION: INDIA

IBMS already inventories 172,517 National Highway structures on the same 0–9 idea, and MoRTH's nationwide digital re-survey is creating exactly the fresh paper baseline this method audits. Morbi — 135 dead, four days after a renovation nobody checked against the physical bridge — is the failure mode, precisely.

Infrastructure does not fail silently.

It fails contradicted. DISSENT is the machine that files the dissent.

THE LIVE CONSOLE

dissent-nexus.netlify.app

THE SOURCE CODE

github.com/ArockiaRajamanickam/dissent

TEAM NEXUS NETWORK

DEPARTMENT OF CSE (CYBER SECURITY)

234,801 REAL FILINGS · 621,137 STRUCTURES MAPPED · NOT ONE INVENTED NUMBER